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High temperature oxidation behaviour of TiAl–Cr–Nb–Mo alloys

Daniela Pilone*, Ferdinando Felli, Andrea Brotzu

Dipartimento ICMA, Sapienza Università di Roma, Via Eudossiana 18, 00184 Roma, Italy

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ABSTRACT

The oxidation behaviour of six different TiAl–Cr–Nb–Mo alloys was investigated by thermogravimetric method. Isothermal experiments were carried out at 900 °C and 1000 °C in air.

The aim of this work was to evaluate the effect of aluminium and of other alloying elements on the oxidation behaviour of TiAl intermetallic alloys. The alloy's microstructure and composition as well as the composition distribution of the oxide scales were analysed by Scanning Electron Microscopy (SEM), Energy Dispersive Spectroscopy (EDS) and X-Ray Diffractometry (XRD). The study highlighted that the oxidation resistance of the considered TiAl-based alloys is strongly affected by its aluminium content, while it seems that an increase of Cr, Nb and Mo concentrations does not affect the oxidation behaviour of the alloy. Yttrium addition, at least in the tested quantity, has a detrimental effect since it promotes the formation of a TiO_2–Al_2O_3 mixed oxide instead of a compact and thin alumina layer.

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1. Introduction

Due to their attractive properties, γ-TiAl based alloys are interesting for high-temperature applications in aerospace and automotive industries. Their potential is seen in low density, high specific yield strength, high specific stiffness, good oxidation resistance at room temperature (RT), resistance against "titanium fire", and good creep properties up to high temperatures [1]. In fact the good specific mechanical properties of titanium aluminide alloys push the development of these materials.

Because of their ordered structure, intermetallics have high mechanical strength both at RT and at high temperature [2,3]. Despite that, TiAl-based alloys cannot be used as single phase alloys since they have a very low ductility at RT. The most interesting alloys are called near gamma. The presence of a second α_2 phase (Ti_3Al) allows control of the microstructure. As far as mechanical properties are concerned, the addition of ternary elements such as Cr, V and Mn reduces the grain size with consequent ductility improvement. Depending on alloy chemistry and microstructure, these alloys exhibit good workability, medium-to-good tensile properties, tensile fracture strains in the range 1–3% at RT and fracture toughness values in the range 10–25 MPa$\sqrt{m}$ [4–7].

To date, various TiAl-based alloys have been developed. Adding transition metals of a high melting temperatures is generally beneficial to increasing the high temperature strength of these alloys [8–10]. Over the last decade the so-called 2nd and 3rd generation alloys have been developed in order to improve mechanical properties and high temperature capabilities [1] of those alloys. The 4th generation alloys, also called "air-hardenable", have been intensively studied as potential materials for investment casting of low pressure turbine blades [11].

Unfortunately poor oxidation resistance currently limits the use of untreated TiAl above 800 °C. When these alloys are subjected to oxidation in air the scale is not a protective Al_2O_3 layer, but a mixture of Al_2O_3 and TiO_2. Their oxidation resistance is known to be significantly affected by the addition of ternary elements [12–17]. In fact improvements in the oxidation resistance of gamma alloys have been achieved by modifying the alloy compositions, particularly by the addition of Mo, W and Nb [8,12,17]. There have been various theories proposed to account for these improvements, but there is considerable support to the hypothesis that the "doping effect" [10] determines the suppression of TiO_2 growth. Many authors have postulated that in TiO_2 formed during oxidation, vacancies play a role in inward oxygen diffusion [18].

When ions with valence +5 or +6, such as Nb and W, are incorporated in the scale, oxygen vacancy concentration should be reduced and this presumably results in the suppression of TiO_2 growth. Recent research [19,20] on high Nb intermetallics highlighted that a dopant element with a higher valence than that of titanium is expected to decrease the oxygen vacancy concentration owing to the electroneutrality in the oxide, and thus to suppress TiO_2 growth. In addition, the $(Ti,Nb)O_2$-rich layer represents a dense and chemically uniform scale. Other studies showed that the combination of Nb and Si can improve the oxidation resistance of

* Corresponding author. Tel.: +39 0644585879; fax: +39 0644585641.
E-mail address: daniela.pilone@uniroma1.it (D. Pilone).

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the Ti–Al based alloys. Si changes the typical microstructure of the oxide scale by promoting the formation of a compact Al_2O_3 in the inner part of the oxidation scale. When Si content is higher than 3% the Ti_5Si_3 phase forms and coarse TiO_2 crystals form on Ti_5Si_3 phase during oxidation. By increasing Nb content in the TiAl-based alloys it is possible to suppress Ti_5Si_3 phase formation and thus a careful balance between Nb and Si content avoids TiO_2 growth [21].

2. Experimental procedure

The alloys used in this work (Table 1) were produced by induction melting under an Ar atmosphere from pure Ti, Al, Cr, Nb, Mo and Y. The molten metal was cast directly into the rotating mould. Isothermal oxidation experiments were carried out at 900 and 1000 °C in static laboratory air. The specimens used in the oxidation tests were polished up to 600 grit SiC papers and cleaned with acetone before oxidation. The weight of the specimen being oxidised was measured by using a Cahan microbalance in conjunction with a computer.

In order to perform metallographic examinations, test specimens prior to and after oxidation tests were ground to a mirror-like surface using SiC papers up to 1200 followed by 0.3 μm alumina. Specimens needing cross-section inspections were first embedded in a cold mounting resin. Metallographic structure, scale morphology and specimen cross sections were inspected by scanning electron microscope (SEM) and microanalyses were carried out by energy dispersion spectroscopy (EDS). Reaction products formed during the oxidation process were examined by X-ray diffraction (XRD) and SEM equipped with EDS.

3. Results and discussion

In this work the oxidation behaviour of TiAl intermetallics was studied on alloys having the as-cast structure. In fact in TiAl-based alloys different microstructures can be obtained through solid-state phase transformations during continuous cooling. Those microstructures can be the final microstructures in components, such as the lamellar microstructure, or a precursor to the final microstructures, such as the massive gamma microstructure. This massive transformation is a diffusionless transformation in which a highly faulted gamma is generated from the parent high-temperature alpha phase by appropriate rapid cooling. It is also well known that some ternary additives to TiAl-based alloys support the massive transformation even at relatively low cooling rates [20]. In this work the alloys were produced by means of centrifugal casting. Because of the relatively high cooling rates involved in the process and of the high alloying element content the resulting microstructures are constituted by fine γ grains. The alloys, which contain a high quantity of β stabilisers, such as Cr, Nb and Mo, show also the presence of β phase (Fig. 1) at the grain boundaries where β stabilisers segregate during solidification. Fig. 1 shows the typical microstructure obtained for the tested alloys.

Specimens of the 6 considered alloys were subjected to isothermal oxidation tests. Fig. 2 shows thermogravimetric oxidation curves obtained at 900 °C in air. The results shown in Fig. 2 indicate clearly that at 900 °C the mean oxidation kinetics are much higher for the alloys characterised by a lower aluminium content. The weight gain after 70 h reaches 14.6 mg cm^{-2} for Alloy 1 and 13.8 mg cm^{-2} for Alloy 2. Fig. 2 highlights also that by increasing the alloy aluminium content the oxidation kinetics are

Table 1
Chemical composition (%at.) of the tested alloys.

	Ti	Al	Cr	Nb	Mo	Y
Alloy 1	57.0	35.6	4.0	1.0	2.4	0.0
Alloy 2	56.8	37.7	3.5	0.5	1.5	0.0
Alloy 3	48.5	46.0	3.0	0.5	2.0	0.0
Alloy 4	40.1	46.3	5.1	3.1	5.5	0.0
Alloy 5	32.2	48.6	7.2	4.8	7.2	0.0
Alloy 6	48.1	46.0	3.0	0.5	2.0	0.4

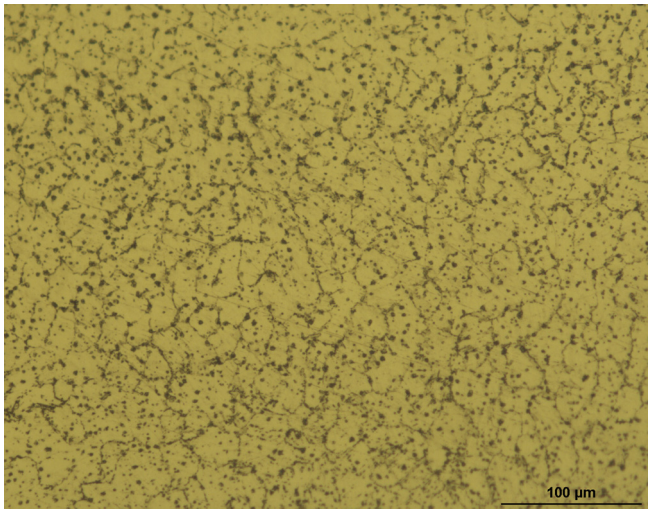


Fig. 1. SEM micrograph showing the microstructure of Alloy 1.

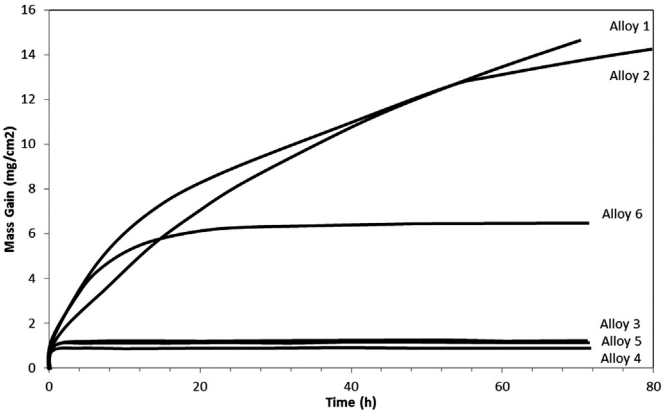


Fig. 2. Isothermal oxidation kinetics for the tested alloys at 900 °C.

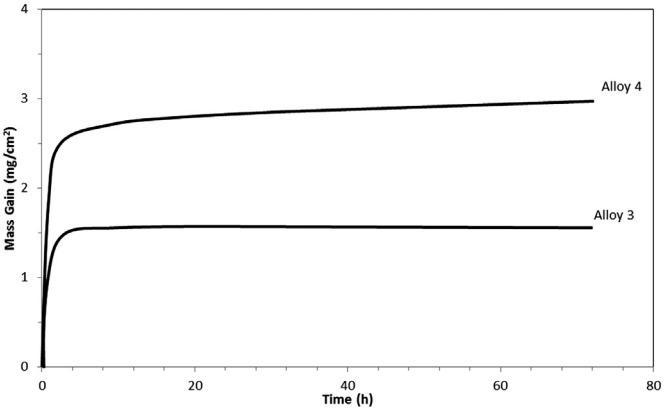


Fig. 3. Isothermal oxidation kinetics for Alloy 3 and Alloy 4 at 1000 °C.

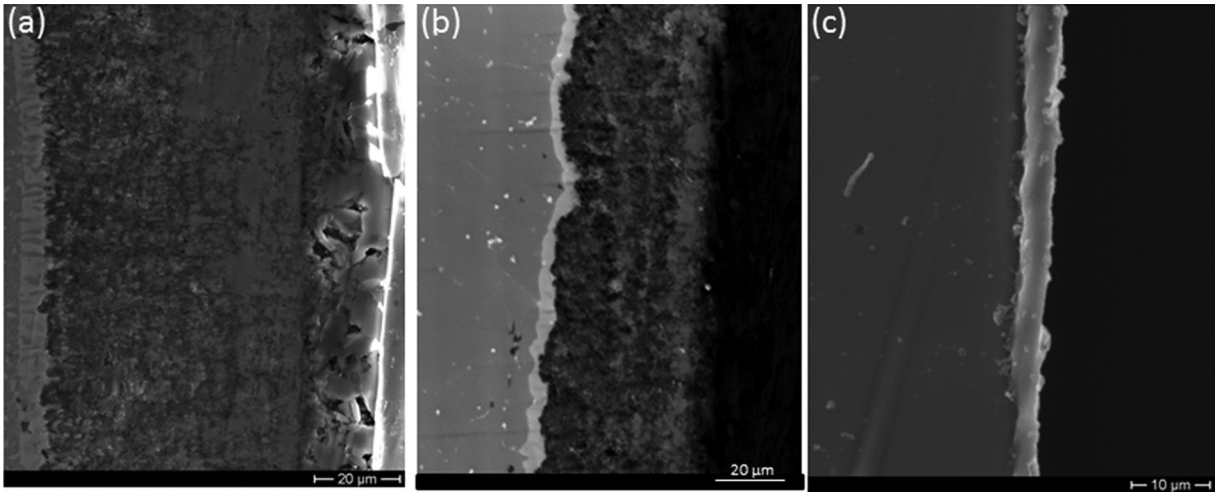


Fig. 4. SEM micrograph of Alloy 1 (a), Alloy 6 (b) and Alloy 5 (c) transverse sections after isothermal oxidation at 900 °C.

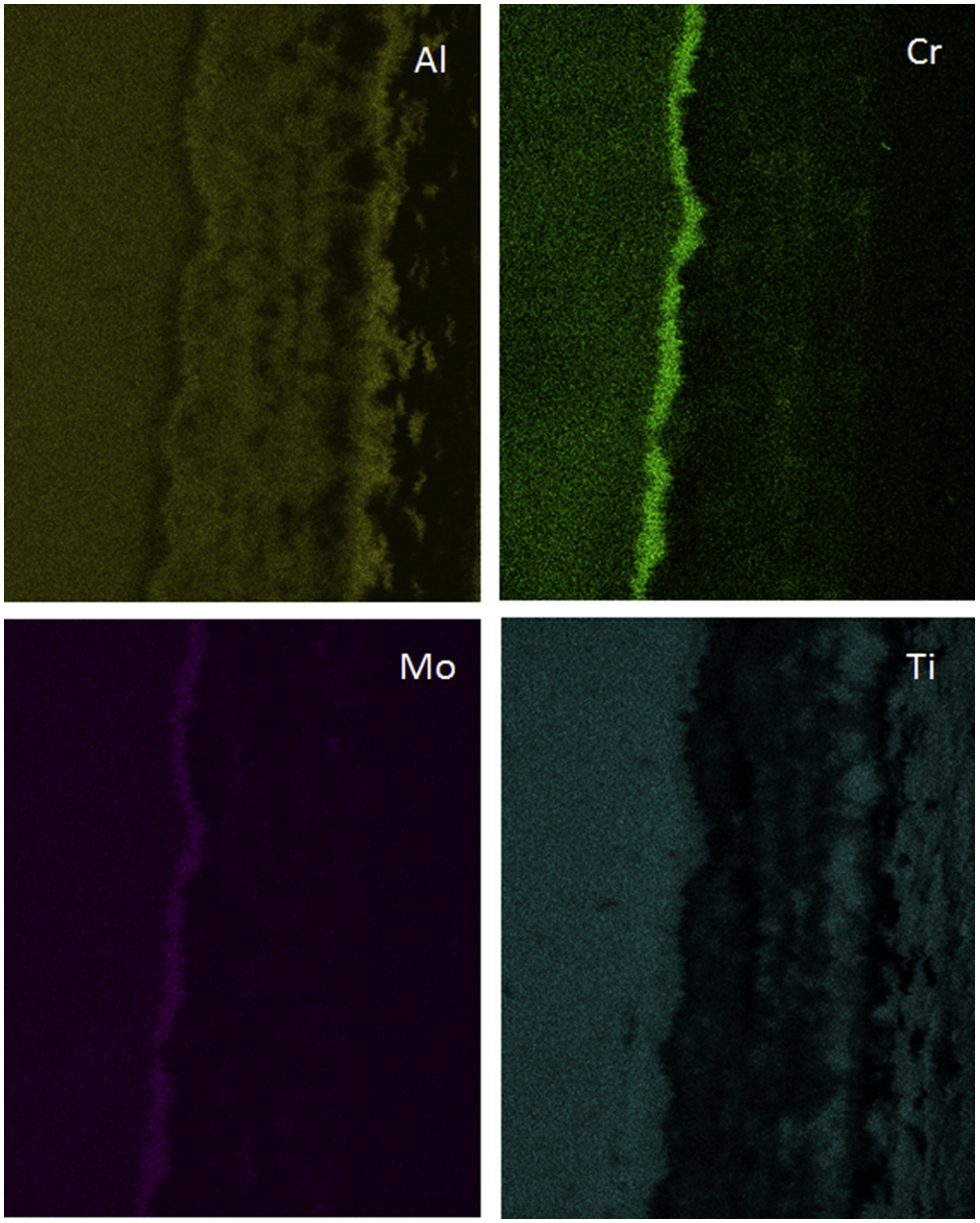


Fig. 5. X-ray maps highlighting Al, Cr, Mo, Ti distribution on the specimen transverse section shown in Fig. 4(b).

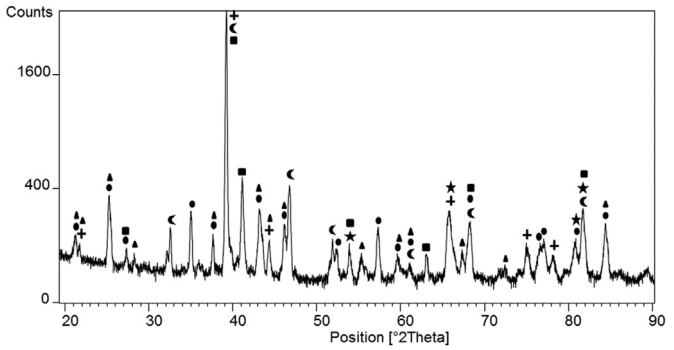


Fig. 6. X-ray diffractograms of Alloy 5 after isothermal oxidation at 900 °C. ✚ TiAl, ☾ $TiAl_3$, ● Al_2O_3, ▲ Nb_2O_5, ★ MoO_2, ■ TiO_2.

lower and very similar at least for Alloy 3, Alloy 4 and Alloy 5. The weight gain of these alloys is about 1.0 mg cm^{-2} after 70 h. The yttrium containing alloy (Alloy 6) shows an intermediate behaviour, in fact during the oxidation test it promotes the formation of an adherent oxide layer that does not spall off during cooling, but the weight gain after 70 h reaches a value of 6.5 mg cm^{-2}. This is probably due to the presence of yttrium that has a very strong affinity for oxygen and thus it promotes oxide scale formation on the alloy surface. In fact it is well known that small additions to bulk alloys of oxygen reactive elements such as Y, Hf, Zr, dramatically improve the adhesion of Al_2O_3 scales [22,23], although the mechanisms producing this effect are still a matter of discussion. A previous work [24] on this topic suggested that a small amount of Y promotes the growth of Al_2O_3 scale but suppresses the development of TiO_2 in the initial stage during isothermal oxidation at 900 °C, while an excessive Y addition assists the development of TiO_2. The effect of an excessive Y addition on the oxidation resistance might be explained by the enhancement of the diffusion of titanium and oxygen due to the increase in lattice defects at 900 °C, resulting in a rapid oxidation.

Considering that the alloys that contain the higher quantity of aluminium and that do not contain yttrium have a very good and similar oxidation behaviour at 900 °C, Alloy 3 and Alloy 4 were tested at 1000 °C. Thermogravimetric oxidation curves obtained at

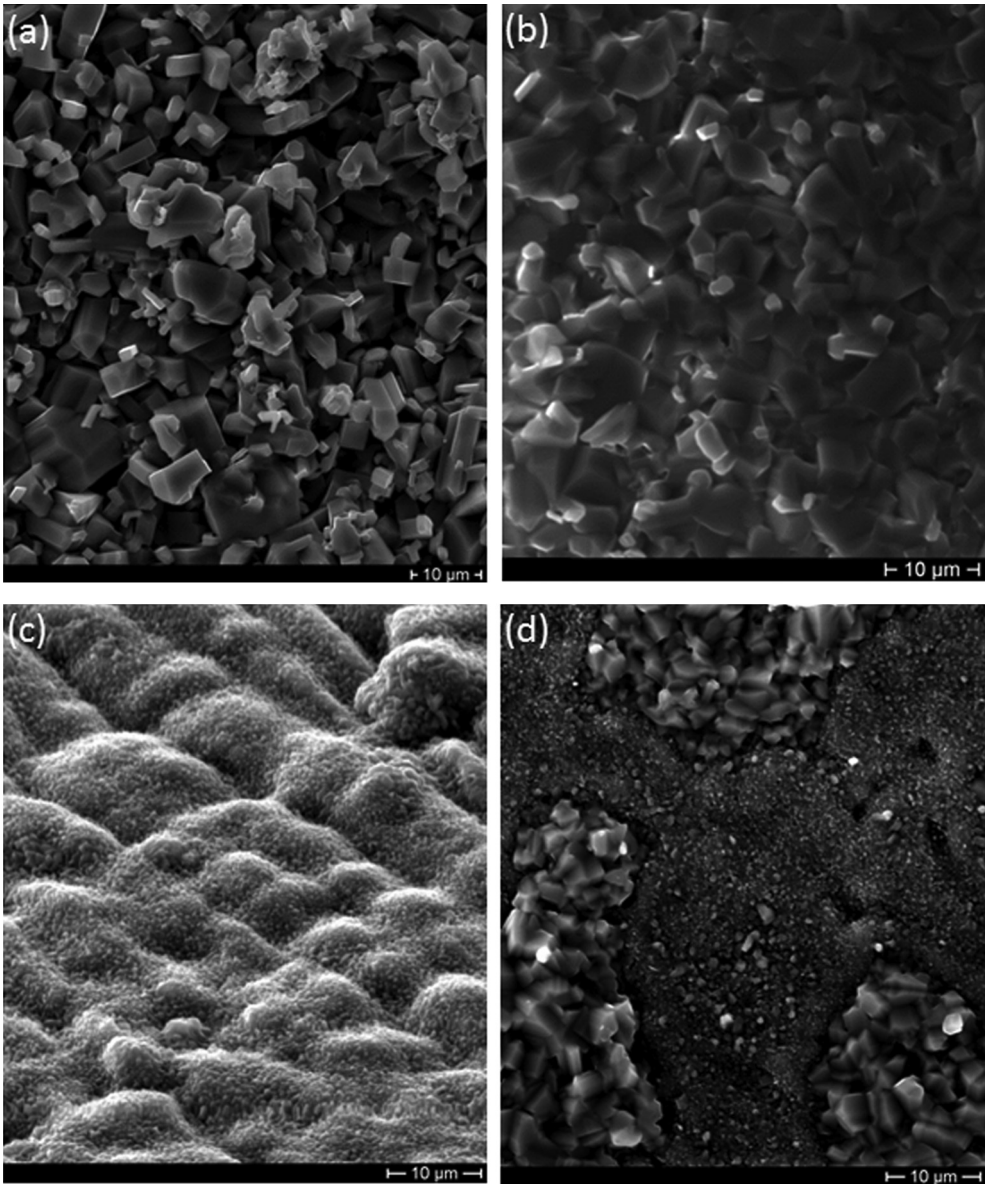


Fig. 7. SEM micrographs showing oxide morphologies of Alloy 2 (a), Alloy 6 (b), Alloy 4 (c) and Alloy 3 (d) after isothermal oxidation at 900 °C.

1000 °C showed again that both alloys have a protective behaviour and a rather limited weight gain (Fig. 3). After 70 h Alloy 3 shows a weight gain of about 1.5 mg cm^{-2} and Alloy 4 a weight gain of about 2.8 mg cm^{-2}.

The knowledge of overall oxidation kinetics is crucial in studying the high temperature oxidation commonly controlled by the diffusion of reactive species through the oxide scale and/or in the subjacent alloy. Diffusion-driven oxidation usually produces parabolic kinetics described through the parabolic law $\Delta m^2 = k_p t$, where Δm is the mass gain per unit area at time t and k_p is the parabolic rate constant. As far as the best behaving alloys is concerned, at 900 °C the experimental curves obey the parabolic law only in the first 20 min since after that time interval the oxide layer becomes very protective and the weight gain remains almost constant. The calculated k_p value is about $5.27 \cdot 10^{-4}$ mg^2 cm^{-4} s^{-1} for Alloy 3, $10.35 \cdot 10^{-4}$ mg^2 cm^{-4} s^{-1} for Alloy 4 and $13.64 \cdot 10^{-4}$ mg^2 cm^{-4} s^{-1} for Alloy 5. The kinetics were reasonably reproducible and represented for these alloys a very protective behaviour.

After oxidation tests samples were cut to inspect by SEM the transverse section with the aim of studying oxide composition and metal-oxide interface. Examination in cross section of specimens treated at 900 °C revealed that Alloy 1 and Alloy 2 oxide scales have a very similar morphology and thickness. As it can be observed in Fig. 4(a) the scale is about 70 μm thick. EDS analyses revealed that the primary scale constituents are alumina and titanium oxide. X-ray mapping showed also that in these specimens a chromium rich layer was formed just below the oxide scale (light grey phase in Fig. 4(a)), but it was not able to promote formation and stabilisation of a protective Al_2O_3 layer. The back-scattered image shown in Fig. 4(a) also reveals that the oxide scale consists of alternating layers of Al_2O_3 and TiO_2. This intermixed scale is not protective and tends to spall off during cooling.

Alloy 6 showed a superior oxidation resistance. The micrograph in Fig. 4(b) and in Fig. 5 show that the scale thickness is about 40 μm and that the oxide layer consists of Al, Ti, Mo and Cr oxides. While a small quantity of Cr and Mo oxides seem to be uniformly distributed throughout the oxide layer, TiO_2 is mainly located in the oxide outermost part. It is impossible for Al_2O_3 to form since the outermost layer consists of TiO_2 and that is because TiO_2 is growing much faster compared to Al_2O_3 [25,26]. As a consequence of the slower Al diffusion, an Al_2O_3 rich layer, is formed under the TiO_2 layer. Since TiO_2 may have defective structure, the inward and outward diffusion of ions occurs. X-ray mapping shows also that a rich bright Chromium–Molybdenum phase is formed in the sub-

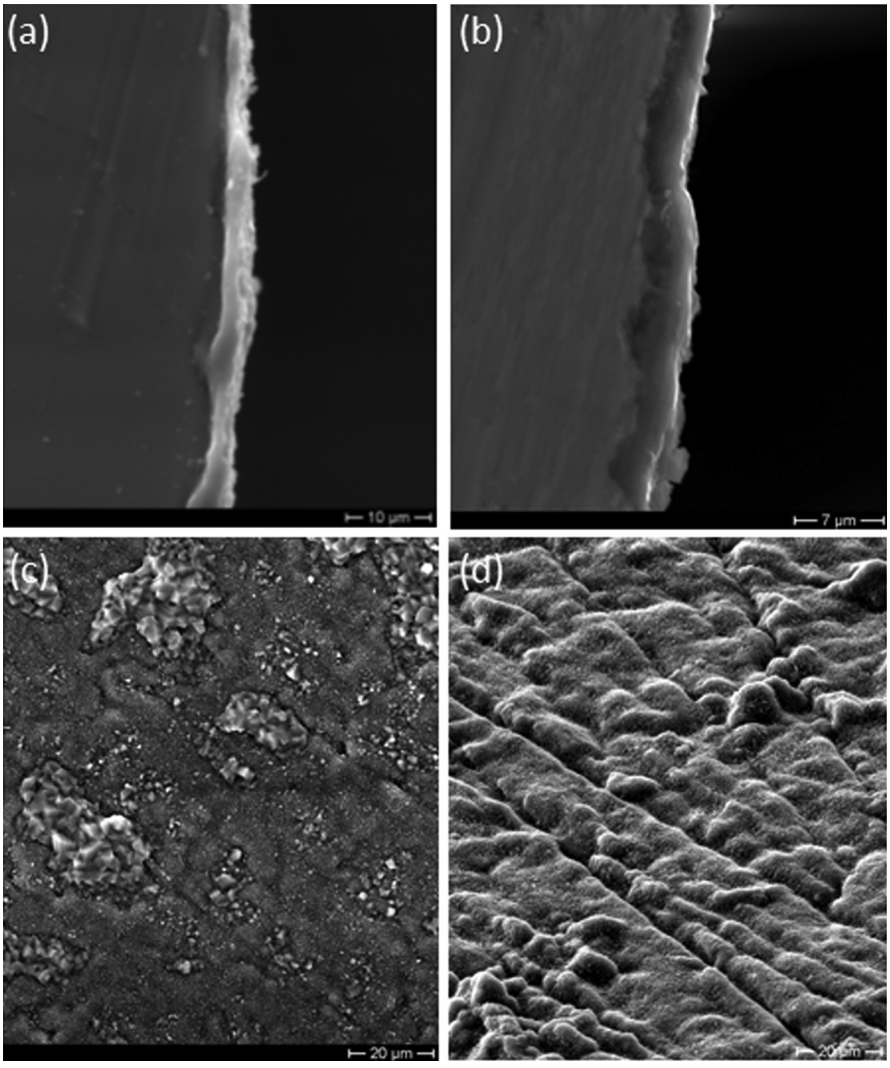


Fig. 8. SEM micrographs showing transverse sections of Alloy 3 (a) and Alloy 4 (b) and oxide morphologies of Alloy 3 (c) and Alloy 4 (d) after isothermal oxidation at 1000 °C.

scale area. Chromium and Molybdenum do not diffuse concurrently with Al and Ti in the outward direction from the bulk alloy, but remain predominantly in the metallic substrate, thus increasing their concentration.

Alloy 3, Alloy 4 and Alloy 5 exhibit a very protective behaviour due to the formation of a thin, compact and adherent oxide scale (Fig. 4(c)). Considering that the oxide scale of these alloys is few micrometres thick it was difficult to determine the oxide composition by means of either EDS analyses or X-ray mapping. The different phases present in the oxide scale were then identified by means of X-ray diffraction. The pattern reported in Fig. 6 shows that the analysed scale is not a pure alumina scale, but it contains also small quantities of Nb, Mo and Ti oxides. Several theories have been proposed in literature for explaining the effect of Nb and Mo addition on the oxidation behaviour of TiAl base alloys [27,28]. Our findings can be explained considering that molybdenum reduces the oxygen solubility in the alloy, causing a transition from internal to external oxidation. Moreover incorporation of molybdenum and niobium into the rutile lattice results in a reduction of defect concentration. The subsequent slow growth of rutile favours the establishment of a thin and adherent alumina layer that can act as a barrier. The results of the present study highlighted that the lowest tested concentrations of Mo and Nb are sufficient to produce the doping effect. Due to doping there would be more time for the Al_2O_3 particles to grow and interconnect with each other, promoting the formation of a self-healing protecting layer.

As far as Cr is concerned, it is usually added to TiAl base alloys to enhance their plasticity [29]. It is reported [30,31] that chromium at low concentration (<4 at.%) has a detrimental effect on the oxidation resistance of TiAl intermetallics and that at high concentration (>10 at.%) has a beneficial effect. In fact during oxidation in air TiAl alloys with high chromium content promote the formation, just below the oxide scale, of a $Ti(CrAl)_2$ Laves phase layer, which is capable of continuous Al_2O_3 scale formation. In this study Cr remains predominantly in the metallic substrate and does not seem to significantly affect the oxidation behaviour of the tested alloys.

In order to understand the oxidation behaviour of the studied alloys it is important to analyse the oxide morphology (Fig. 7). Alloy 2 exhibits the formation on the external surface of the typical TiO_2 polyhedric crystals (Fig. 7(a)), while on the surface of Alloy 6 polyhedron edges are blunt and the oxide seems to be more compact (Fig. 7(b)).

Alloy 4, which contains a higher quantity of Mo, Nb and Cr in comparison with Alloy 3, shows an outermost alumina layer that has a very fine structure characterised by ripples and globular morphology. It is well known that when the alumina scale grows almost exclusively by inward oxygen transport, it appears flat since the new oxide is formed nearly exclusively at the metal/oxide interface [32]. In contrast, a mixed transport mode of aluminium and oxygen determines the formation of new oxide within the scale and then the formation of convolutions.

Alloy 3 and Alloy 5 promote the formation on the surface of a fine alumina layer and, in some isolated areas, titanium oxide rosettes, that do not affect the resistance to spalling, are visible.

In order to gather additional information on the oxidation behaviour of the studied alloys, transverse sections and oxide morphologies of Alloy 3 and Alloy 4 after isothermal oxidation at 1000 °C have been analysed (Fig. 8). Fig. 8 shows that even at 1000 °C the tested alloys promote the formation of a thin, compact and adherent oxide scale. As far as the oxide morphology is concerned, at 1000 °C Alloy 3 and Alloy 4 form oxides with a morphology very similar to the ones obtained at 900 °C. This observation confirms that an increase of Mo and Nb quantity in the alloy reduces defect concentration in the oxide and favours the formation of a self-healing alumina layer that is protective even at 1000 °C.

4. Conclusions

The study carried out in this paper highlighted that the oxidation resistance of TiAl–Cr–Nb–Mo alloys is strongly affected by their aluminium content. The research showed that for the considered alloy compositions the oxidation resistance is poor when the tested alloys contain less than 40% Al, while it is very good when the aluminium content is higher than 45%. It seems also that Cr, Nb and Mo concentration increase slightly affects the oxidation behaviour of the alloy. The addition of yttrium, at least in the tested concentration, has a detrimental effect since it promotes the formation of a TiO_2–Al_2O_3 mixed oxide instead of a compact and thin alumina layer.

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